Natural Language Processing

Assignment- 8

TYPE OF QUESTION: MCQ

Number of Questions: 8 [Q4, Q6 carries 2 marks each] Total Marks: 10

Question 1: Consider the following statements. Which of the following is/are True?

1. Dog is a hyponym of animal.
2. Fruit is a hypernym of apple.
3. Animal is a hyponym of dog.
4. Oak is a hypernym of tree.

Answer: 1, 2

Question 2: Which of the following is False?

1. Hypernym: From specific concepts to more general concepts
2. Hyponym: From general concepts to more specific concepts
3. Holonym: From parts to the wholes they belong to
4. Antonym: Words that have opposite meanings

Answer: 2

Solution:
A hyponym relation goes from a specific concept to a more general concept (e.g., *rose* → *flower*).
Statement 2 incorrectly describes the direction, so it is False.

Question 3: Two concepts along with their glosses are given below. Find the similarity score between concepts "teacher" and "instructor" using the Extended Lesk's algorithm.

(Note: Do not consider the stop words.)

teacher: a person who teaches students in a school

instructor: a person who teaches learners in an educational institution

1. 2
2. 3
3. 5
4. 8

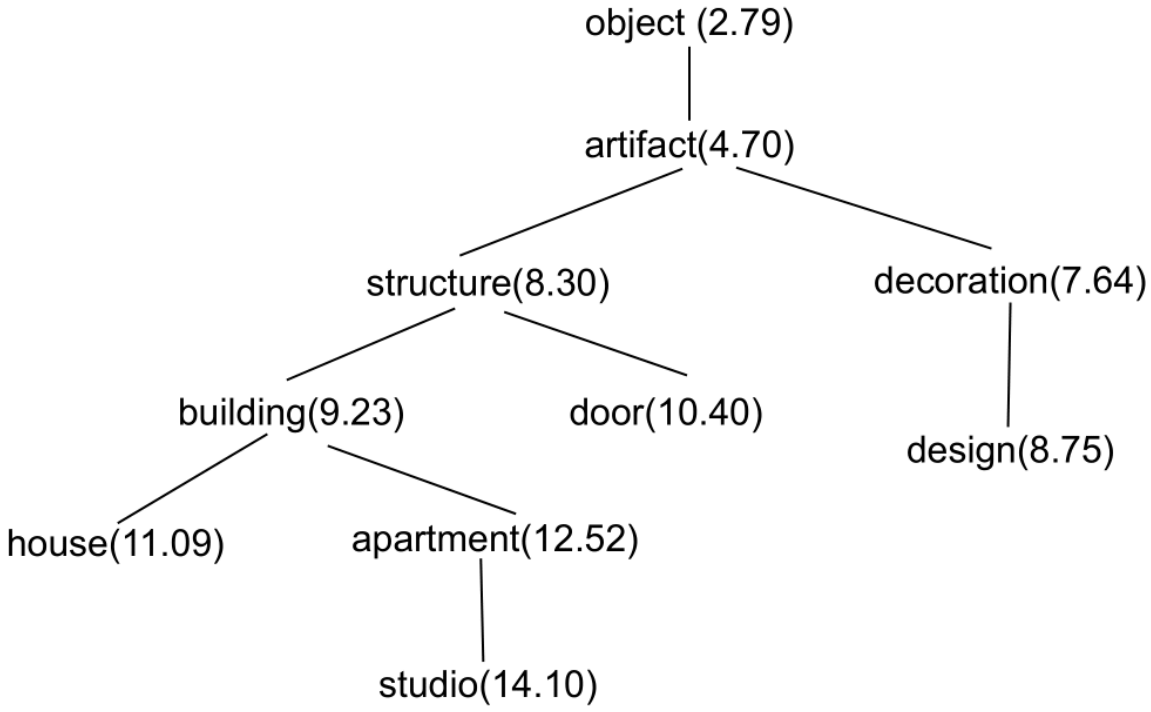
Answer: 1

Solution:

Common words are: **person, teaches**

Score = 1^2 + 1^2 =2

For Question 4 to 6, consider a hypothetical wordnet noun taxonomy with their information content as shown in Figure 1. Question 4 carries 2 marks
Note: Use base 10 in logarithmic calculations



Question 4: What is the Lin similarity between **house** and **decoration**?

a. 0.564
b. 0.533
c. 0.501
d. 0.473

Answer: c
Solution: $(2\times4.7)/(11.09+7.64) \approx 0.501$

Question 5: What is the Resnic similarity between **house** and **apartment**?

a. 11.09
b. 8.30
c. 9.23
d. 4.70

Answer: c

Solution:

Resnik Similarity between "house" and "apartment"

The Resnik similarity measure is based on the information content (IC) of the lowest common ancestor (LCA) in a taxonomy. The LCA is the most specific node that is an ancestor of both terms.
The LCA of "house" and "apartment" is "building" with an IC of 9.23.

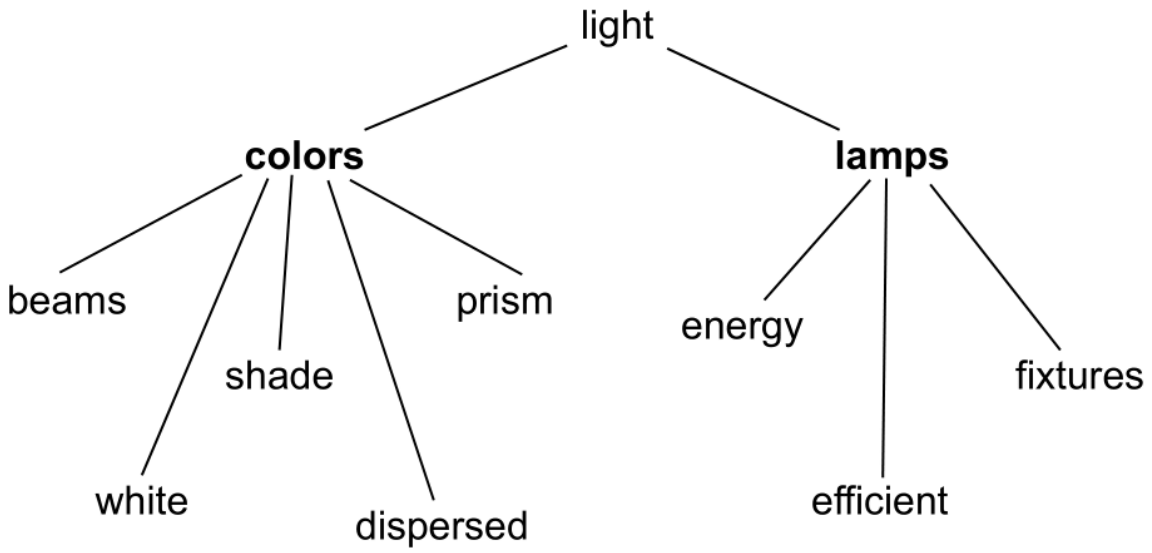
Question 6: What is the Leacock–Chodorow similarity between **building** and **design**?

a. 0.398
b. 0.699
c. 0.097
d. None of the above

Answer: a

Solution:
`LC similarity = −log pathlen(c1,c2)/2d = −log 4/(2×5) ≈ 0.398`

For Question 7 to 9 consider the network of words for disambiguation of the word "light" as shown in Figure 3. The hubs are "colors" and "lamps". Note: Take the distance between two words as the path length between them.



Question 7: Compute the scores for (i) the hub "colors" and the component "white" and (ii) the hub "colors" and the component "fixtures".

a. 0.2, 0.25
b. 1.0, 0.0
c. 0.5, 0.25
d. None of the above

Answer: d
Solution:
(i) 1/(1+1) = 0.5
(ii) 0 as "colors" is not an ancestor of "fixtures"

Question 8: What are the scores of the hubs "colors" and "lamps" respectively?

a. 0.6, 0.4
b. 0.20, 0.33
c. 2.5, 1.5
d. None of the above

Answer: c
Solution: Each component's score is 0.5